Up to 4 billion people rely on herbal medicine.
Most medicinal plants in trade are taken from the wild.
Almost no one is counting what is left.

Harvester Risk Atlas

Which wild medicinal plants of the Philippines need help first, where they grow, and how sure we are.

Yuan Andrei C. Mariano · Alicia Jhyle Vilale



Kalingag (*Cinnamomum mercadoi*), found only in the Philippines. Photo: Nielbert Raner, CC BY 4.0, iNaturalist

Sources: Kew, State of the World's Plants and Fungi 2020 (via Natural History Museum); IUCN / TRAFFIC, *Wild at Home*, 2018.

THE PROBLEM

The world's medicine cabinet is being emptied, and almost no one is checking

30,000

plant species have documented medicinal uses.

60–90%

of those in international trade are **collected from the wild**.

7%

have ever been assessed for extinction risk.

1 in 5

of those assessed are **already threatened**.

Trade in these plants **tripled in value** between 1999 and 2015, to over **US$3 billion**. When roots or bark are taken, the plant often dies. When a medicinal plant disappears, so does a remedy families have used for generations.

Source: IUCN Medicinal Plant Specialist Group and TRAFFIC, *Wild at Home* (Jenkins, Timoshyna & Cornthwaite 2018), iucn.org, 28 June 2018. Root and bark harvest: Cunningham 2001.

THE PROBLEM, CLOSE TO HOME

In the Philippines, the forests these plants need shrink every day

175 ha

of tree cover lost **every day** from 2001 to 2023, about **245 football pitches a day**.

1.47 million ha in total: 7.9% of the tree cover of 2000.

1,186

plants on the government's **threatened list**. 248 are critically endangered.

But harvest is never measured, so no one can say which medicinal plants are being dug up fastest.



Tongkat ali

Its **roots are dug up** and sold as medicine. Officially Endangered. In the whole country, open databases hold only **5 reliable records** of it.

Tree-cover loss (30% canopy; includes plantations): Hansen et al. 2013 / Global Forest Watch. DENR DAO 2026-20, our count. GBIF. Photo: Steven Bodzin, CC BY 4.0.

OUR ANSWER

A free map that ranks which plants need help first, and says why in plain words

For each native medicinal plant we ask three simple questions, using only open data. Each counts equally.

RARITY

How hard is it to find?

How seldom botanists find it where they have looked hard, and how small its world range is.

HARVEST HARM

Does harvesting kill it?

Taking roots, bark or the whole plant scores highest. Slow-growing trees and wild-only plants score higher.

HABITAT PRESSURE

Is its home disappearing?

How much of where it grows is unprotected, and how fast forest is being lost there.

Climate change is shown on each plant's page but not scored: projected warming is about +1.5 °C for every plant, so it cannot tell them apart.

Harvest Priority Index v2 = ⅓ Rarity + ⅓ Harvest harm + ⅓ Habitat pressure. Equal weights are the neutral default (OECD/JRC 2008). Full method: docs/METHODS.md.

DATA AND METHOD

From 9,660 raw records to a ranking we re-test 2,000 times

1 · COLLECT

6

open sources: GBIF records, Kew's plant checklist, protected areas, forest loss, climate, and Stuart's *Philippine Medicinal Plants*.

2 · CLEAN

4,789

reliable records kept out of 9,660, after 9 checks (bad coordinates, garden plants, duplicates and more).

3 · SCORE

29

native plants scored, out of 51. The other 22 are listed but not ranked: not native, or too few records.

4 · STRESS-TEST

2,000

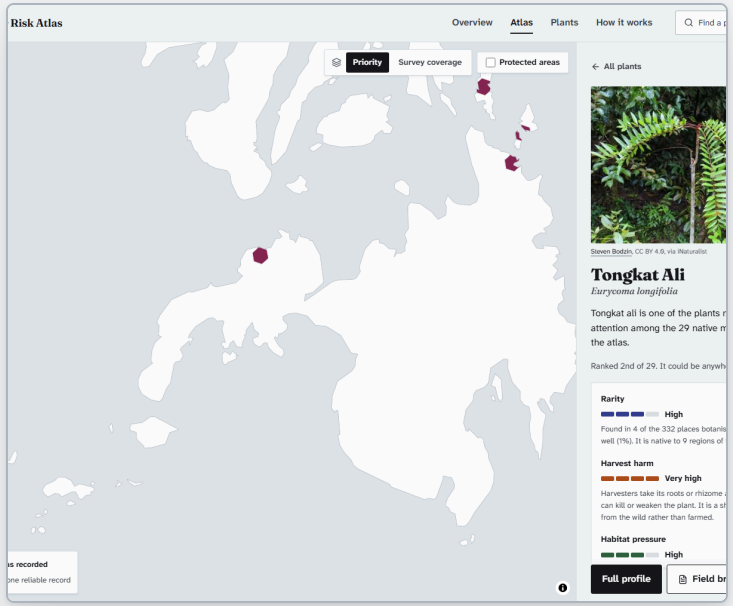
re-runs with slightly different weights and evidence. Each plant gets a rank **range**, not just one number.

Fairer rarity. Busy roadsides get photographed more than remote forest. So we measure rarity only in the **332 well-searched areas** (out of 1,267), where a missing plant really means it is scarce.

H3 hexagons ≈ 250 km². Uncertainty: Dirichlet weights, Jeffreys Beta reporting rate, habitat bootstrap. Rebuild: python -m harvester_risk_atlas all --offline

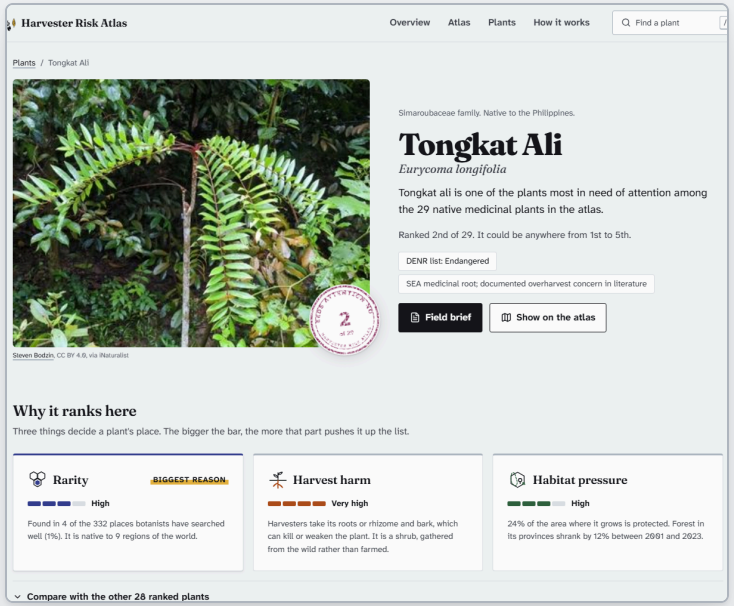
PROTOTYPE · LIVE AT HARVESTER-RISK-ATLAS.VERCEL.APP

See it, understand it, take it to the field



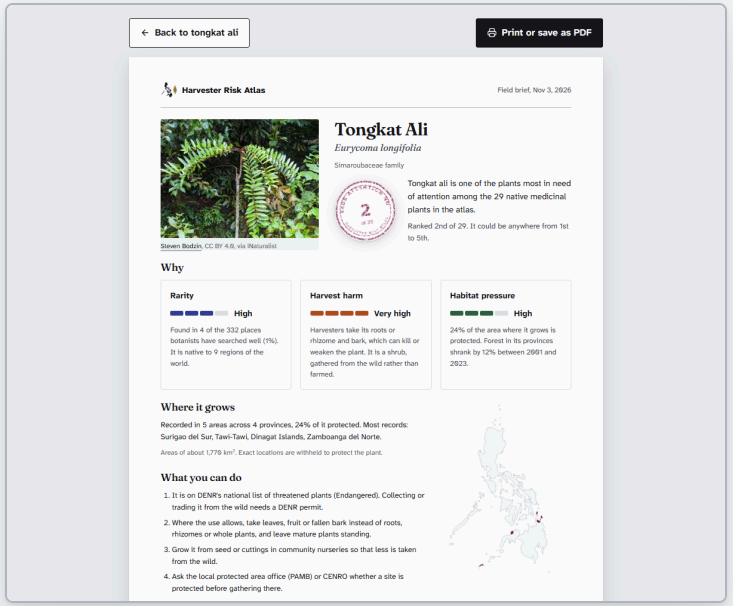
1 · See where

Plants appear only as broad hexagons, never exact spots, so the map can't lead collectors to them.



2 · Understand why

Three reasons in plain words, and how far the rank could move.



3 · Take it to the field

A one-page brief to print, with what rangers and harvesters can do.

Also: a weight explorer to try your own priorities, CSV download, phone layout. React + TypeScript + MapLibre; 73 Python, 66 unit and 37 browser + accessibility tests.

RESULTS

Five plants need attention now



Kalingag

Cinnamomum mercadoi

2nd (could be 1st–3rd)

Found only in the Philippines. Its bark is harvested.



Tongkat ali

Eurycoma longifolia

2nd (could be 1st–5th)

Roots dug up. Officially Endangered.



Bitaog

Calophyllum blancoi

3rd (could be 1st–4th)

A forest tree; its bark and sap are taken.

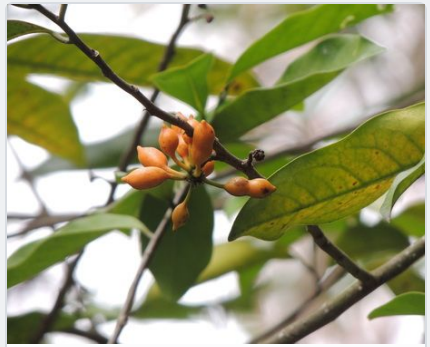


Tibig

Ficus nota

4th (could be 3rd–5th)

A wild fig; bark gathered from the wild.



Amuyon

Goniothalamus amuyon

5th (could be 2nd–9th)

Only 6 reliable records in the country.

For all five, the biggest reason is **rarity**. 4 more plants are on **keep watch** and 20 are lower concern. Two of the five are on DENR's threatened list, even though the list was never an input.

Two plants share median rank 2. Photos via iNaturalist: Nielbert Raner (CC BY), Steven Bodzin (CC BY), 葉子 (CC0), Mark Bulahao (CC BY), shulinchen (CC BY-NC).

WHAT'S NEW · CHECKED AGAINST EXPERT SOURCES

Verified against the experts, and turned into a survey plan

50 of 51

names matched in **World Flora Online**, the backbone EthnoHack recommends; all accepted names. The hybrid *Mentha × piperita* is flagged and not ranked.

Agrees

Co's Digital Flora of the Philippines (built on Leonardo L. Co's work) confirms tongkat ali as native and DENR Endangered. Disagreements are shown, never hidden.

66%

of the land next to known records of the 29 ranked plants has barely been searched: 2,992 of 4,542 neighbouring areas.

Where to look next, on every plant page: places Co's Flora documents with no open record yet (tongkat ali: **Mindoro, Homonhon, Bucas Grande**), under-searched areas beside known records, and records outside the documented range to double-check. And the ranking holds: change the maths and the order barely moves (Spearman ρ = 0.97).

What it is not, and where it can be wrong

We are careful

- **Not medical advice**, not a harvest permit, not a Red List assessment.
- **No exact locations**: only ~250 km² areas, to protect plants from collectors.
- **Traditional knowledge** comes only from a published, cited source. No community or personal data collected.
- **Expert review pending**: the harvest traits come from cited sources but **none is expert-checked yet**, and every plant page says so.

Where it can be wrong

- **Records follow people.** We correct for survey effort, but only partly.
- **Harvest harm comes from books**, not field counts of what is taken.
- **Few records, big doubt.** Tongkat ali rests on just 5 records, which is why we show its range.
- **Some results surprise.** Lagundi is listed as introduced, so it is not ranked. We explain rather than override.

WHAT HAPPENS NEXT

Use it today. Help us make it certain.

Who can act on it now

- **DENR, protected-area boards, local governments:** know which plants and areas to survey first.
- **Harvester groups and herbal makers:** see which plants to grow instead of dig up.
- **Researchers and students:** an open, reproducible starting point.

What we will build next

- **Expert checks** with Filipino ethnobotanists and harvester communities.
- **Range models** that predict where plants grow beyond known records.
- **Trade data and more countries**: the global sources already carry over.

harvester-risk-atlas.vercel.app github.com/yuan05-afk/harvester-risk-atlas